

Processing report for CHAN-009

Photogrammetry data processing report for model "Chinese rython" (CHAN-009)
for the 'Chiriandreses Museum' Project.

3D Scanning Technician: Eva Perez Chirinos.

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Survey Data

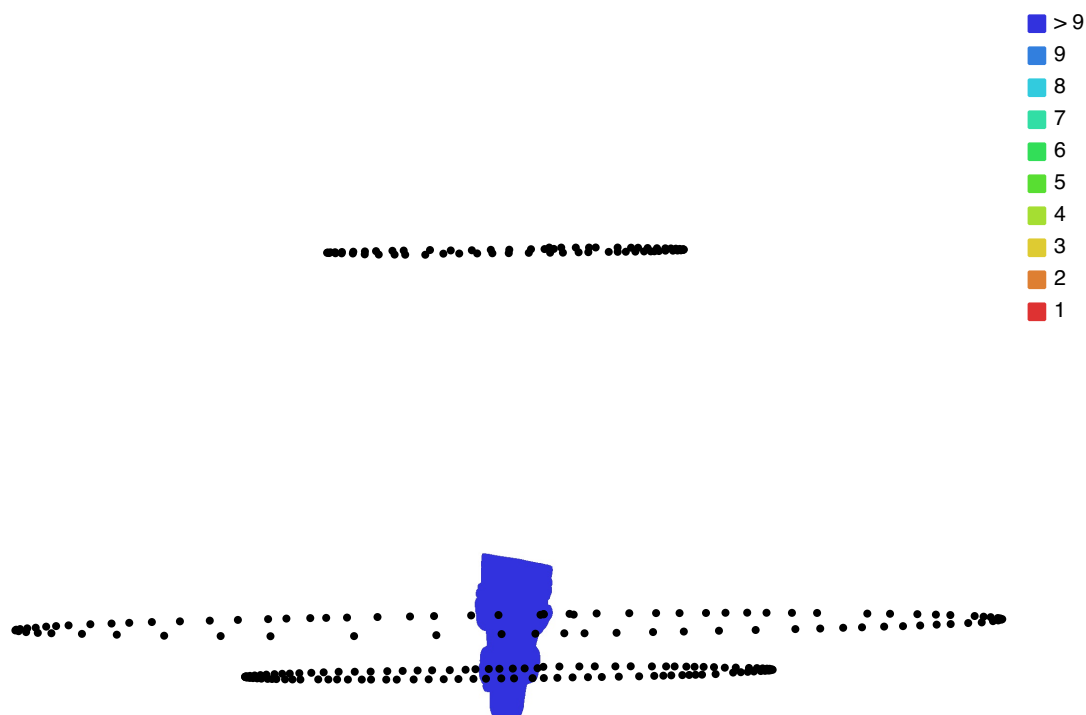


Fig. 1. Camera locations and image overlap.

Number of images:	256	Camera stations:	256
		Tie points:	350,988
		Projections:	1,137,964
		Reprojection error:	0.595 pix

Camera Model	Resolution	Focal Length	Pixel Size	Precalibrated
Canon EOS 2000D, EF-S1...	6000 x 4000	21 mm	3.77 x 3.77 μ m	No
Canon EOS 2000D, EF-S1...	6000 x 4000	32 mm	3.77 x 3.77 μ m	No
Canon EOS 2000D, EF-S1...	6000 x 4000	25 mm	3.77 x 3.77 μ m	No

Table 1. Cameras.

Camera Calibration

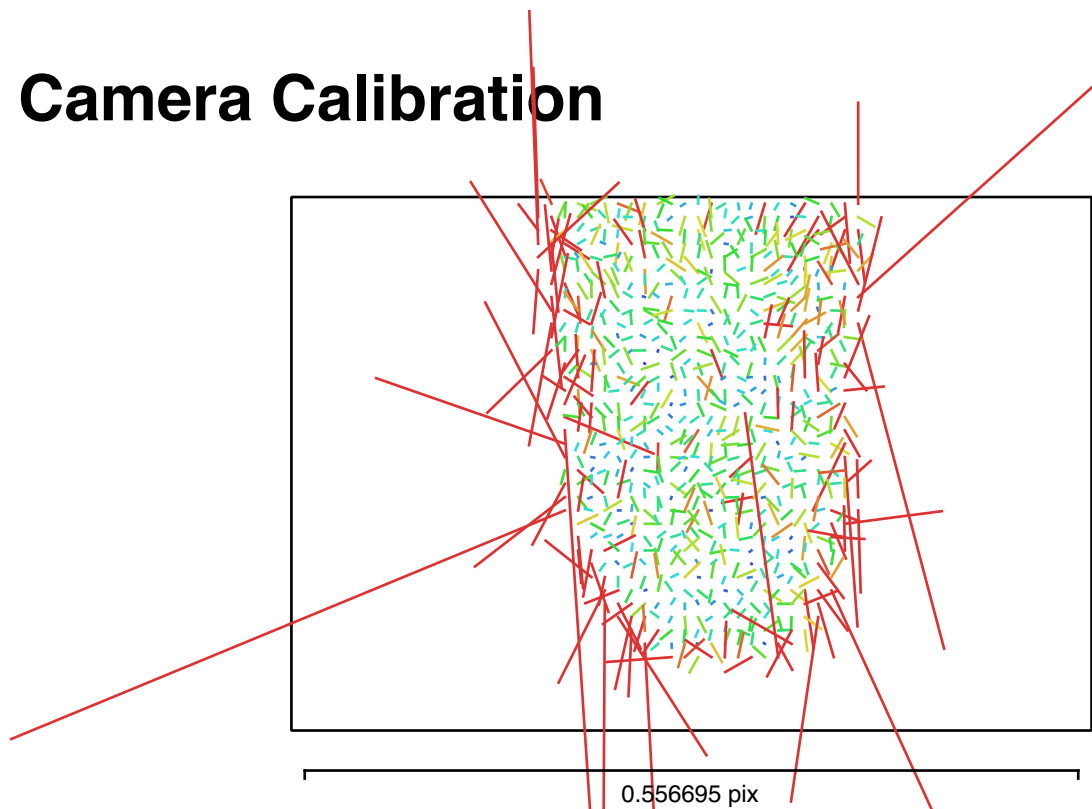


Fig. 2. Image residuals for Canon EOS 2000D, EF-S18-55mm f/3.5-5.6 IS II (21mm).

Canon EOS 2000D, EF-S18-55mm f/3.5-5.6 IS II (21mm)

104 images

Type	Resolution	Focal Length	Pixel Size
Frame	6000 x 4000	21 mm	3.77 x 3.77 μm

	Value	Error	F	Cx	Cy	B1	B2	K1	P1	P2
F	5519.41	0.85	1.00	-0.12	-0.31	-0.88	0.26	-0.21	0.08	0.18
Cx	-17.1615	0.18		1.00	0.02	0.08	-0.24	0.07	0.08	0.07
Cy	-37.2429	0.4			1.00	0.41	-0.02	-0.17	0.05	-0.25
B1	7.66858	0.79				1.00	-0.23	-0.14	0.06	-0.39
B2	-2.4285	0.097					1.00	-0.09	0.41	-0.26
K1	-0.141056	0.00027						1.00	-0.39	0.38
P1	0.000659793	1.6e-05							1.00	-0.07
P2	-0.00172197	6.5e-05								1.00

Table 2. Calibration coefficients and correlation matrix.

Camera Calibration

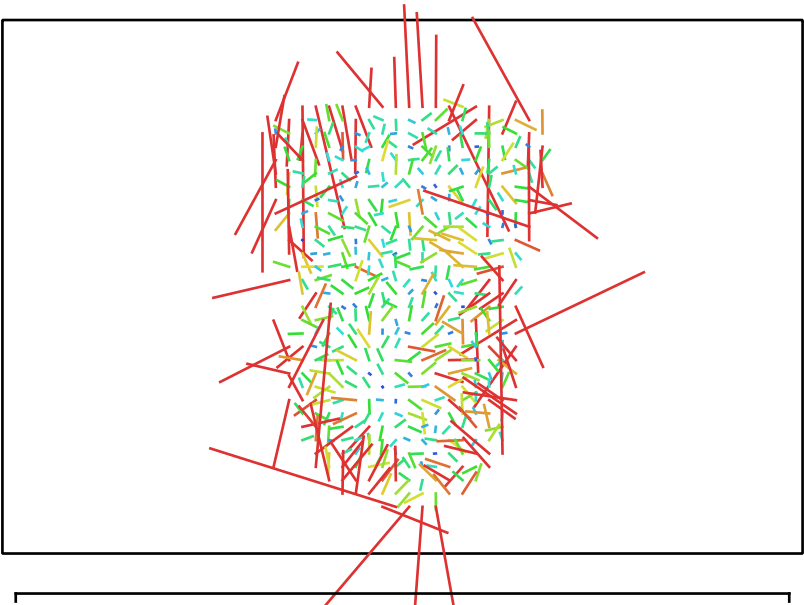


Fig. 3. Image residuals for Canon EOS 2000D, EF-S18-55mm f/3.5-5.6 IS II (32mm).

Canon EOS 2000D, EF-S18-55mm f/3.5-5.6 IS II (32mm)

85 images

Type	Resolution	Focal Length	Pixel Size
Frame	6000 x 4000	32 mm	3.77 x 3.77 μm

	Value	Error	F	B1	B2	K1	P1	P2
F	8372.8	1.6	1.00	-0.68	-0.12	-0.29	0.08	0.23
B1	6.54154	1.2		1.00	0.14	-0.24	0.01	-0.06
B2	-1.64618	0.13			1.00	-0.03	0.20	-0.12
K1	-0.0547576	0.001				1.00	-0.09	-0.12
P1	0.00126242	2.8e-05					1.00	0.02
P2	-0.00265719	0.00012						1.00

Table 3. Calibration coefficients and correlation matrix.

Camera Calibration

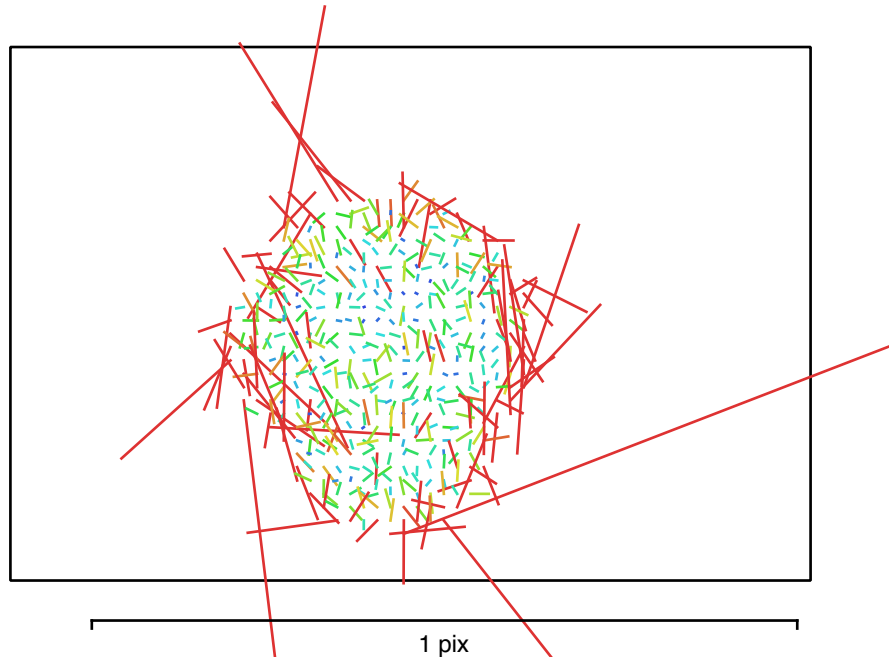


Fig. 4. Image residuals for Canon EOS 2000D, EF-S18-55mm f/3.5-5.6 IS II (25mm).

Canon EOS 2000D, EF-S18-55mm f/3.5-5.6 IS II (25mm)

67 images

Type	Resolution	Focal Length	Pixel Size
Frame	6000 x 4000	25 mm	3.77 x 3.77 μm

	Value	Error	K1	P1	P2
F	6626.1				
K1	-0.127593	0.0016	1.00	0.71	-0.69
P1	0.00173375	5e-05		1.00	-0.28
P2	0.000996157	7.6e-05			1.00

Table 4. Calibration coefficients and correlation matrix.

Digital Elevation Model



Fig. 5. Reconstructed digital elevation model.

Processing Parameters

General

Images	256
Aligned images	256

Tie Points

Points	350,988 of 776,504
RMS reprojection error	0.120191 (0.595095 pix)
Max reprojection error	0.279827 (26.7587 pix)
Mean key point size	3.72519 pix
Point colors	3 bands, uint8
Key points	No
Average tie point multiplicity	4.77268

Alignment parameters

Accuracy	High
Generic preselection	Yes
Reference preselection	No
Key point limit	40,000
Key point limit per Mpx	1,000
Tie point limit	20,000
Filter points by mask	Yes
Mask tie points	No
Exclude stationary tie points	No
Guided image matching	No
Adaptive camera model fitting	Yes
Matching time	4 minutes 47 seconds
Matching memory usage	157.12 MB
Alignment time	4 minutes 47 seconds
Alignment memory usage	307.91 MB

Optimization parameters

Parameters	f, b1, b2, cx, cy, k1, p1, p2
Adaptive camera model fitting	Yes
Exclude corners	No
Optimization time	2 seconds
Date created	2026:04:24 15:21:26
Software version	2.2.2.21287
File size	66.84 MB

Depth Maps

Count	256
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Depth maps generation parameters

Quality	High
Filtering mode	Mild
Max neighbors	16
Processing time	25 minutes 35 seconds
Memory usage	2.93 GB
Date created	2026:04:24 16:18:48
Software version	2.2.2.21287
File size	428.24 MB

Model

Faces	200,000
Vertices	100,000
Vertex colors	3 bands, uint8
Texture	8,192 x 8,192, 3 bands, uint8

Depth maps generation parameters

Quality	High
Filtering mode	Mild
Max neighbors	16
Processing time	25 minutes 35 seconds
Memory usage	2.93 GB

Reconstruction parameters

Surface type	Arbitrary
Source data	Depth maps
Interpolation	Disabled
Strict volumetric masks	No
Processing time	2 minutes 50 seconds
Memory usage	1.67 GB

Texturing parameters

Mapping mode	Generic
Blending mode	Mosaic
Texture size	8,192
Page count	1
Enable hole filling	Yes
Enable ghosting filter	Yes
UV mapping time	21 seconds
UV mapping memory usage	138.44 MB
Blending time	21 minutes 30 seconds
Blending memory usage	5.70 GB
Date created	2026:04:24 16:21:48
Software version	2.2.2.21287
File size	105.58 MB

System

Software name	Agisoft Metashape Standard
Software version	2.2.2 build 21287
OS	Mac OS 64 bit
RAM	16.00 GB
CPU	
GPU(s)	Apple M2